

Deep Learning & AI in Sports

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Dr. Pegah Rahimian

Dr. Pegah Rahimian is a football data scientist at Twelve Football and a post-doctoral researcher at Uppsala University. She holds a PhD in Football Data Science and has applied AI to derive in-depth analyses from football data, influencing various aspects of the sport.

October 2025

Mondays & Thursdays:
17:00 - 19:00 CET

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Professor David Sumpter

Professor David Sumpter is a professor of applied mathematics at Uppsala University and the author of "Soccermatics," a book that explores the application of mathematical modeling to football. He co-founded Twelve Football and has worked with major clubs and national federations worldwide.



Course Roadmap

The key is to move from **structure** → **interactions** → **valuation** → **communication**,



Session 1: Tracking → Team Shapes, defensive lines & Formation Clusters



Session 2: Playing-Style & Role Representations (clustering/embeddings)



Session 3: Graph Neural Networks for Pass Risk & Reward, Pitch Control



Session 4: Action Valuation I — Expected Threat (xT), EPV & VAEP



Session 5: Action Valuation II — Plus-Minus & Reinforcement Learning



Session 6: Interpretability & Explainable ML in Football



Session 7: Communicating With LLM Assistants (coach-ready insights)



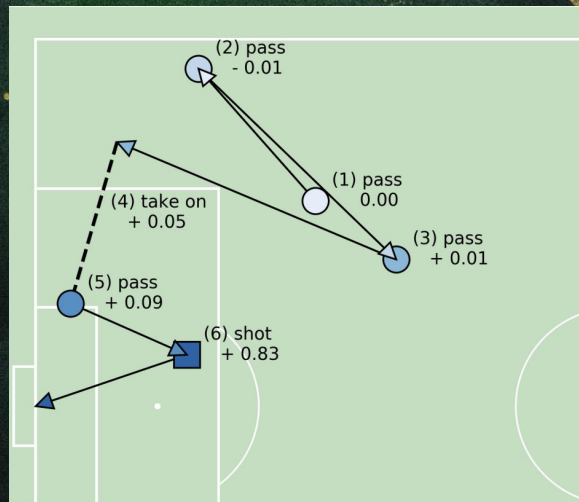
Session 8: Project Studio & Demos

Module 4:

From Possessions to Probability: Understanding xT, VAEP, OBV and Modern Football Action Valuation

Valuing Every Action: The Science Behind Modern Football Analytics

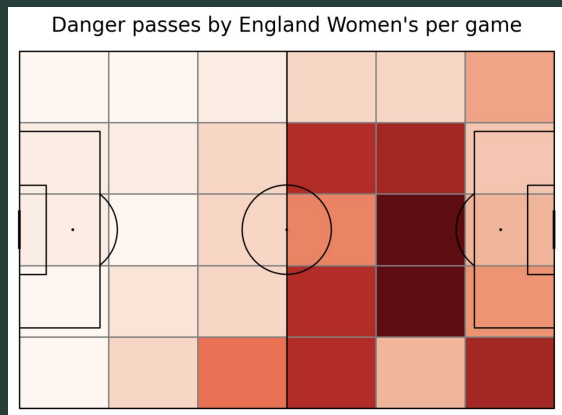
From Expected Threat to VAEP — Turning Events into Insights



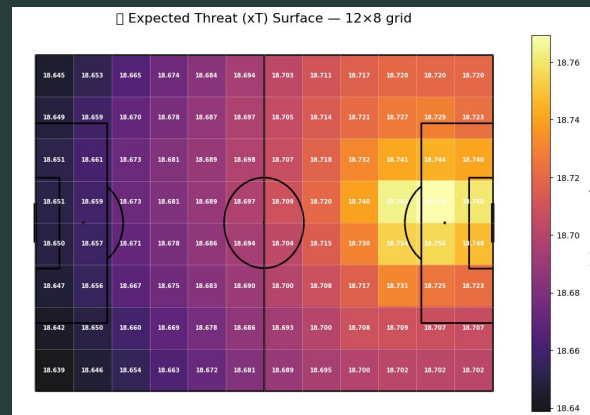
"Every pass. Every decision.
Every bit of **value**."

Why Value Football Actions?

- Football is continuous — not all actions end in goals.
- Goals are rare (~1 every 70 possessions).
- Traditional metrics (passes, shots, assists) miss **contextual impact**.
- We need to **measure contribution**: how each action *changes the team's probability to score or concede*.



Number of passes in each zone leading to goal



Probability of pass starting from each zone leading to goal

Two Perspectives on Value: Where vs What

Football analytics defines “value” in two fundamentally different ways —
where the ball is (position-based) and **what the player does (action-based)**.

Location defines danger

Examples:

- Expected Threat (xT)
- Expected Possession Value (EPV)
- Expected Goals (xG) (at shot moment)



“Value is in
where you are.”

Intuition:

- Every pitch zone has an expected scoring potential.
- Value increases as the ball moves toward goal.

Strengths:

- ✓ Simple, spatially interpretable
- ✓ Great for team-level analysis and tactical mapping

Limitations:

- ✗ Ignores *context* (pressure, teammates, type of action)
- ✗ Can't evaluate decision quality

Actions define contribution

Examples:

- VAEP (Decroos et al., 2019)
- OBV (StatsBomb)
- G+ (Karunaratne, 2022)

Intuition:

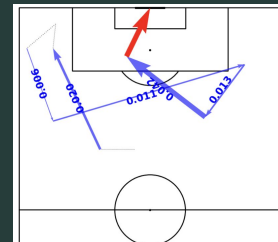
- Value = how much an action improves or harms team success.
- A pass, dribble, clearance, or interception all get credit (or blame).

Strengths:

- ✓ Captures decision quality
- ✓ Works for any on-ball event
- ✓ Great for player valuation

Limitations:

- ✗ Needs labeled data (shots/goals)
- ✗ Harder to interpret visually



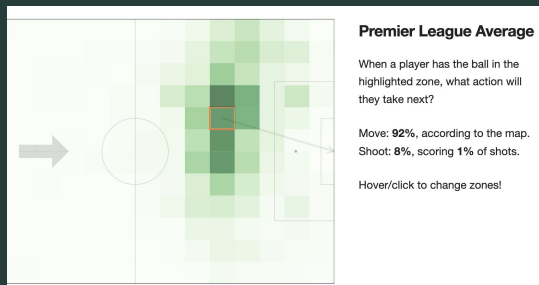
“Value is in
what you do.”

The Evolution of Football Possession Valuation

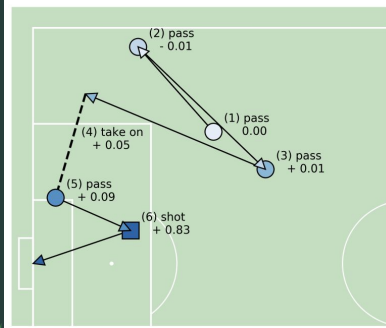
xT

VAEP

OBV



	TIME	PLAYER	ACTION	P_{scores}	VALUE
1	92m4s	S. Busquets	pass	0.03	0.00
2	92m6s	L. Messi	pass	0.02	- 0.01
3	92m8s	S. Busquets	pass	0.03	+ 0.01
4	92m11s	L. Messi	take on	0.08	+ 0.05
5	92m12s	L. Messi	pass	0.17	+ 0.09
6	92m14s	A. Vidal	shot	1.00	+ 0.83



Expected Threat (xT) — Spatial Value

How dangerous is the ball's current location?

- Each cell on the pitch has an *expected threat value* = probability that a possession starting here eventually leads to a goal.
- Computed using a **Markov chain**:

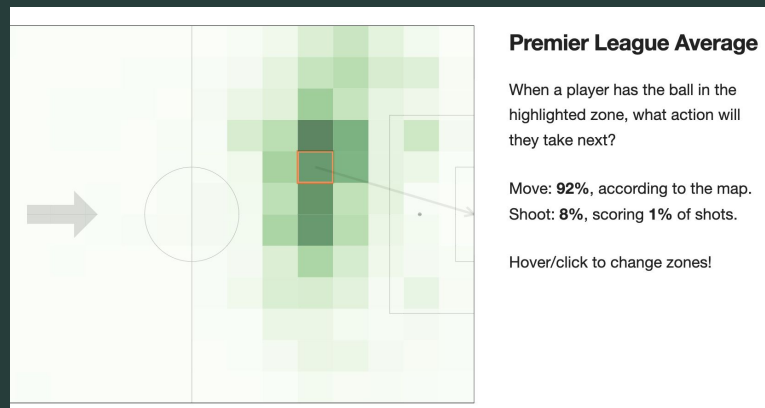
$$\mathbf{xT}_{x,y} = (s_{x,y} \times g_{x,y}) + (m_{x,y} \times \sum_{z=1}^{16} \sum_{w=1}^{12} T_{(x,y) \rightarrow (z,w)} \mathbf{xT}_{z,w})$$

Strengths:

- ✓ Purely spatial, very interpretable
- ✓ Great for tactical analysis — where on the pitch teams generate threat

Limitations:

- ✗ Ignores context (pressure, teammates, speed, etc.)
- ✗ Doesn't distinguish between *types* of actions (e.g., dribble vs pass)
- ✗ Not player- or team-specific — only location-based



VAEP (Valuing Actions by Estimating Probabilities) — Probabilistic Value

How much does an action change the team's probability to score or concede in the near future?

- Uses **supervised ML models** to estimate those probabilities.
- Each state (s_i) includes features like:
 - action type, x/y location
 - distance/angle to goal
 - pressure, teammates nearby, direction of play
- Models: typically **logistic regression**, **XGBoost**, or **neural nets**

Strengths:

- ✓ Captures *context* and *intention*, not just position
- ✓ Provides player-level value per action
- ✓ Differentiates between creating chances and risky play

Limitations:

- ✗ Needs well-labeled data (shot/goal labels)
- ✗ Doesn't account for *off-ball contributions* (e.g., runs, space creation)

VAEP Calculation

$$\Delta P_{scores}(a_i, x) = P_{scores}(S_i, x) - P_{scores}(S_{i-1}, x).$$

$$\Delta P_{concedes}(a_i, x) = P_{concedes}(S_i, x) - P_{concedes}(S_{i-1}, x).$$

$$V(a_i, x) = \Delta P_{scores}(a_i, x) + (-\Delta P_{concedes}(a_i, x))$$

Compute the difference between the values of the pre-action and post-action game states

Pre-action game state



Likelihood to **score** a goal = 0.03

Likelihood to **concede** a goal = 0.02

Post-action game state



Score increase = 0.02

Concede decrease = 0.01

Contribution rating = **0.03**

Likelihood to **score** a goal = 0.05

Likelihood to **concede** a goal = 0.01

Action Valuation = Δ Probability (Score - Concede)

$$VAEP(a_i) = [P(\text{score}|s_{i+1}) - P(\text{score}|s_i)] + [P(\text{concede}|s_i) - P(\text{concede}|s_{i+1})]$$

Value = how much an action improves your situation (or worsens the opponent's)

Positive VAEP → improves chance of scoring

Negative VAEP → risky, leads to danger

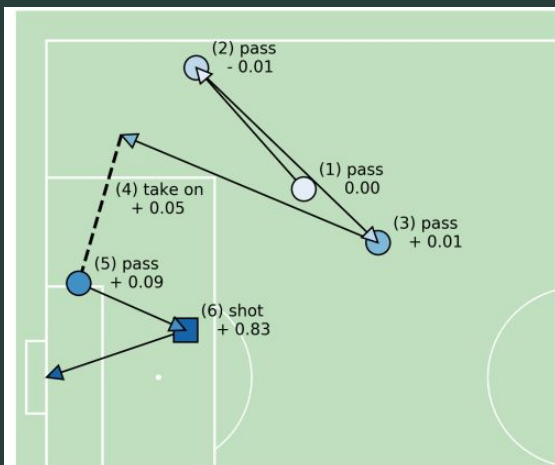


Figure 1: The attack leading up to Barcelona's final goal in their 3-0 win against Real Madrid on December 23, 2017.

	TIME	PLAYER	ACTION	P_{scores}	VALUE
○	1 92m4s	S. Busquets	pass	0.03	0.00
○	2 92m6s	L. Messi	pass	0.02	- 0.01
●	3 92m8s	S. Busquets	pass	0.03	+ 0.01
--	4 92m11s	L. Messi	take on	0.08	+ 0.05
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■	6 92m14s	A. Vidal	shot	1.00	+ 0.83

VAEP Limitation

The two probabilities are modeled jointly from the same features and state, so disentangling *offensive vs defensive credit* is tricky (a defensive action might indirectly help attack).

OBV (On-Ball Value) — Generalized Value Model

OBV (On-Ball Value): Measuring Every Action's Impact on Expected Goal Difference

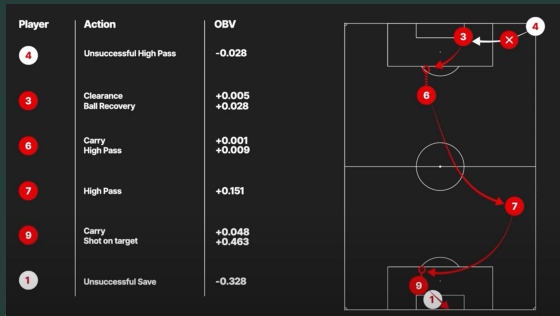
- Same underlying principle ($\Delta P(\text{score}) - \Delta P(\text{concede})$)
But StatsBomb trains **two separate, dedicated models**:
 - one for “Goals For” probability (xGF)
 - one for “Goals Against” probability (xGA)
- This allows:
 - richer, asymmetric feature sets for attack vs defense
(e.g., pressure, line-breaking passes for attack; compactness, interceptions for defense)
 - separate interpretability for *attacking* and *defensive* contribution

“OBV extends VAEP by unifying offensive and defensive value — measuring how every touch affects the game’s expected outcome.”
— *StatsBomb, 2021*

So, mathematically:

$$OBV(a_i) = [xGF_{i+1} - xGF_i] - [xGA_{i+1} - xGA_i]$$

where xGF and xGA come from *two independently trained classifiers*.



What “OBV extends VAEP” really means

When StatsBomb says:

“OBV extends VAEP by unifying offensive and defensive value...”

They mean **conceptually**, OBV generalizes the VAEP logic:

- It combines both offensive and defensive dimensions **into one unified metric** of expected goal *difference* (rather than reporting separate “offensive VAEP” and “defensive VAEP” scores).
- It explicitly decomposes each into **two independent probability models**, giving more modeling flexibility and richer features for both phases.

In other words:

- **VAEP**: “How valuable was this action?”
(joint scoring and conceding view)
- **OBV**: “How did this action change our expected *goal difference*?”
(explicit xGF – xGA balance, with separate models)

***But how to estimate the
scoring and conceding
probabilities***

?

From Possession to State

What is a “state”?

A *state* summarizes what's going on before or after each action:

- Location
- Pressure
- Direction of play
- Teammate and opponent context



Possession segmentation: sequences of actions until team loses control

Probability Estimation



The Labeling Problem:

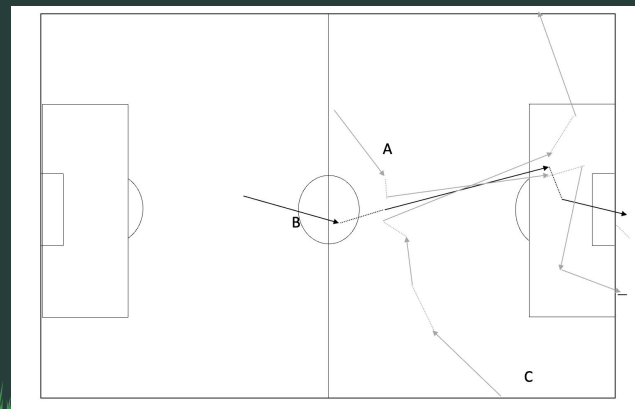
Teaching the Model What “Threat” Looks Like

We label each state with:

- **Timing:** Will there be a **shot/goal** within X seconds?
- **Counting:** Will there be a **shot/goal** within X actions?
- **Possession-based:** Will there be a goal at the end of the possession

Features:

- Distance, angle, progress, x/y coordinates
- Forward/backward movement
- Under pressure, teammates support



Feature Importance

Logit Regression Results

```

=====
Dep. Variable:    goal_10a    No. Observations:    362853
Model:            Logit      Df Residuals:          362841
Method:           MLE       Df Model:              11
Date:            Sun, 19 Oct 2025    Pseudo R-squ.:        0.01675
Time:            23:32:28    Log-Likelihood:        -34717.
converged:        True       LL-Null:               -35309.
Covariance Type:  nonrobust    LLR p-value:           7.586e-247
=====
  
```

```

=====
              coef    std err          z      P>|z|      [0.025      0.975]
-----
const          -3.9805      0.013    -310.904      0.000     -4.006     -3.955
x_start_rescaled -0.6503      0.113     -5.744      0.000     -0.872     -0.428
y_start_rescaled -0.0909      0.057     -1.590      0.112     -0.203      0.021
x_end_rescaled   0.3202      0.102      3.141      0.002      0.120      0.520
y_end_rescaled   0.0429      0.052      0.823      0.410     -0.059      0.145
distance          0.0291      0.029      1.011      0.312     -0.027      0.085
progress_y       -0.0360      0.020     -1.836      0.066     -0.074      0.002
angle_to_goal_start -0.0139      0.024     -0.583      0.560     -0.060      0.033
distance_to_goal_start -0.6169      0.066     -9.383      0.000     -0.746     -0.488
is_forward        0.0711      0.015      4.849      0.000      0.042      0.100
pressure_context  -0.1231      0.014     -8.816      0.000     -0.150     -0.096
teammates_support  0.0728      0.013      5.648      0.000      0.048      0.098
=====
  
```

	Shot_Coeff	Goal_Coeff	Shot_pval	Goal_pval
Feature				
const	-1.826031	-3.980476	0.000000	0.000000
x_start_rescaled	-0.599690	-0.650349	0.000000	0.000000
x_end_rescaled	0.598292	0.320155	0.000000	0.001681
distance_to_goal_start	-0.471306	-0.616853	0.000000	0.000000
teammates_support	0.101650	0.072787	0.000000	0.000000
pressure_context	-0.060994	-0.123112	0.000000	0.000000
is_forward	0.054113	0.071095	0.000000	0.000001
distance	-0.032343	0.029071	0.005900	0.312179
y_start_rescaled	-0.032113	-0.090865	0.157018	0.111925
y_end_rescaled	0.017257	0.042896	0.402753	0.410294
progress_y	0.014189	-0.035975	0.076278	0.066352
angle_to_goal_start	-0.006857	-0.013862	0.477107	0.559992

The Valuation Family Tree

Model	Core Concept	Typical Use	Type
xT	Spatial threat potential	Tactical & positional play	Markov Chain
VAEP	Action-level Δ score, Δ concede	Player valuation	ML classification
G+ (Goal Impact)	Single-metric scoring contribution	Broadcast-friendly metric	Hybrid
EPV (Expected Possession Value)	Value evolution in tracking	Micro-decisions	RL / Monte Carlo
OBV (On-Ball Value)	Value of all touches	Generalized VAEP	ML



From Valuation Models to Football Decisions

Data → Model → Normalization → Interpretation → Application

Turning Action Valuation Models into Scouting & Evaluation Tools

Models like **xT**, **VAEP**, and **OBV** estimate *expected value change per action*.

But raw values are *not directly comparable* across teams, leagues, or positions.

To use them in **scouting or recruitment**, we must:

1. Normalize the metrics
2. Aggregate across time horizons (per 90 mins, per possession)
3. Contextualize by role and tactical system

Goal: Create standardized player/team-level scores that reflect **decision-making quality, threat creation, and defensive risk management**.

$$V_{player} = \frac{1}{N} \sum_i \text{VAEP}(a_i) \quad \text{or} \quad V_{team} = \frac{1}{N} \sum_i \text{xT}(\text{state}_i)$$

How to Standardize xT / VAEP / OBV for Fair Comparison

Category	Method	Example
Per-90 normalization	Divide by player's minutes	$VAEP_per90 = total_VAEP / minutes_played \times 90$
Possession-weighted normalization	Divide by number of team possessions	$OBV_per_possession = total_OBV / team_possessions$
Context normalization	Adjust for game state, scoreline, opponent	Zonal xT
Positional z-scores	Compare within position group	$(z = (x - \mu_{\{pos\}}) / \sigma_{\{pos\}})$
League scaling	Rescale to common baseline (e.g., EPL mean = 0)	$(x' = (x - \mu_{\{league\}}) / \sigma_{\{league\}})$

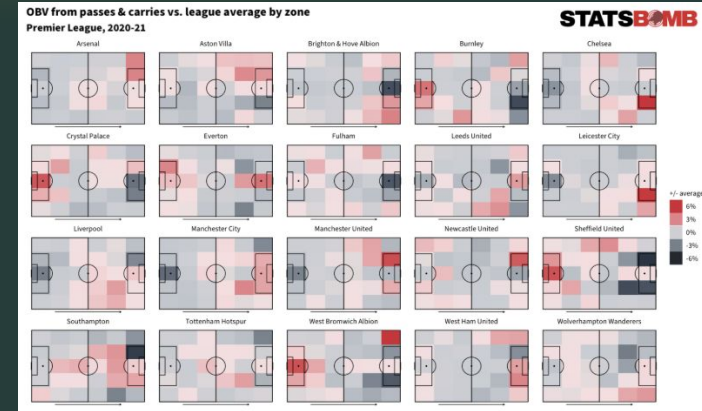
Key insight:

A forward and a full-back should not be compared on raw VAEP or xT, standardization ensures fairness and interpretability.

Team-Level Aggregation and Style Fit

- Aggregate xT, VAEP, OBV **per possession, per zone, or per tactical phase.**
- Cluster teams on aggregated patterns → identify *playing styles* and *system fit*.

Metric	Example aggregation	Interpretation
xT	Mean xT per entry zone	Territory-based attacking structure
VAEP	Average VAEP per possession	Team decision-making quality
OBV	Net OBV balance (xGF – xGA)	Expected goal differential per 90

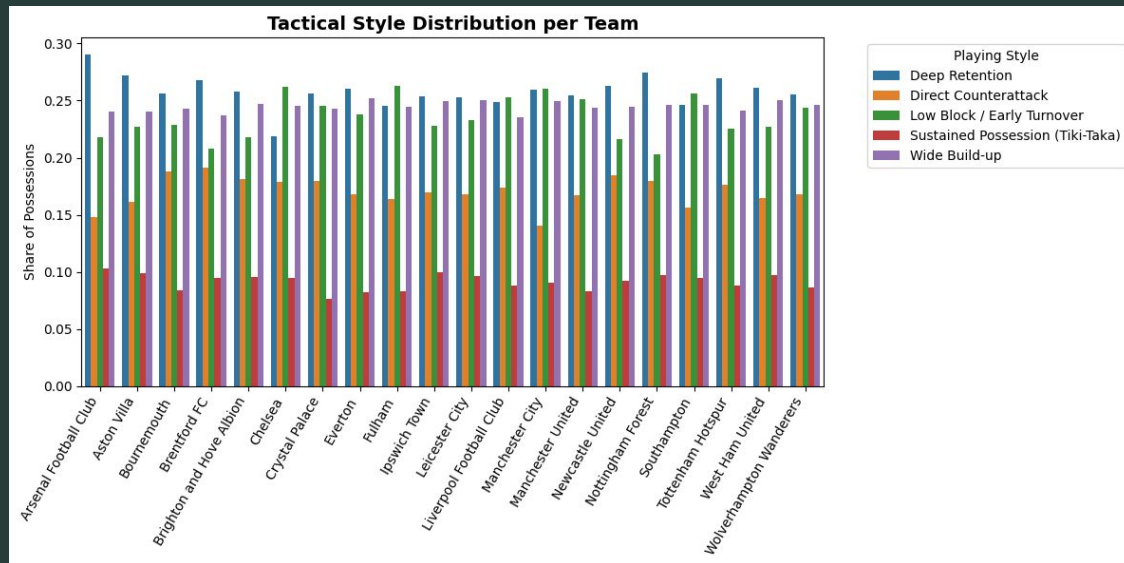


Application:

Recruitment analysts can measure *fit*:

- “This midfielder’s VAEP profile matches high-possession systems.”
- “This defender’s OBV (defensive component) excels in high-pressing structures.”

Applying to Scouting and Recruitment



Player-Team fit analysis

During playing style analysis, we detected Arsenal excelling at Deep Retention possession chains.

1. We aggregate the players xT/VAEP/OBV generation across all games.
2. The players with highest value generated in Deep Retention possessions are fit to Arsenal playing style.

Research & Engineering Challenges in Action Valuation

Area	Project Idea	Scientific Angle
Modeling	Build context-aware VAEP (include pressure, line height, game state)	Multimodal learning (event + tracking)
Cross-league scaling	Normalize xT/VAEP across leagues (e.g., EPL vs Serie A)	Domain adaptation & rescaling
Team-fit simulation	Build “what-if” OBV engine for transfers	Causal inference / counterfactual reasoning
Explainability	Apply SHAP or TabNet to VAEP/OBV outputs	Feature contribution visualization
Spatiotemporal modeling	Replace static features with transformers over possession sequences	Sequential modeling in MDPs